

the type of the component—resistor, capacitor or inductor—and indicate its value with accuracy.

Standard LCR meters come in shapes similar to that of multimeters, with cable probes connected to them. These meters are not very comfortable to work with SMD components. You cannot even touch and hold a 0603 package with these probes. But precise gold-plated tweezers are able to pick even the smallest SMD components and take measurements from already mounted devices.

Most toolkits do not include a smart tweezer but consider buying one to complete your toolkit. When buying a smart tweezer, look for below mentioned features that are found useful in electronics jobs:

Automatic identification. With this feature, smart tweezers automatically determine the type of the component—resistor, capacitor or inductor. This is really a handy feature when working with SMD components. It usually happens that several loose SMD components end up on your table after a project. You cannot distinguish these components easily through naked eyes and identifying the value of each component is almost impossible. If you simply pick and hold the component with a smart tweezer, it will automatically show the component type together with its accurate value.

Automatic ranging. This is another useful feature where the device automatically selects a proper range and signal frequency for highest accuracy measurements. So you do not have to set the range every time you want to measure the value of a component. Manual overrides should also be available because sometimes you would need to see the measurements with different number of significant figures.

Hold mode. This is a highly useful feature for taking measurements at those places on the boards where you cannot see the display of smart tweezers. This feature holds the measured value on the display, so you can see it comfortably.

Continuity/open test, diode polarity testing. Smart tweezers that come

Addendum:

ERSA I-con Pico is a good-quality soldering station that was missed in the buyers' guide on temperature-controlled soldering stations published in April issue.

Features and specifications	ERSA I-con Pico
	
Analogue/digital	Digital
Wattage	
Overall	80 W
Soldering pencil	80 W
Temperature range	150°C to 450°C
Temperature stability	± 1°C at idle temperature
Tip to ground potential	< 2 mV
Tip to ground resistance	< 2 ohms
Thermal recovery	6 seconds
Input/output voltage	220-240 V/50 Hz
Heating element	Nickel chromium-sensortronic-based
Iron stand	Yes
Dimensions	145 x 80 x 103 mm
Cord length	1.2 m, heat-resistant
Standard tip	Chisel-shaped, 1.6 mm
Cleaning wire	Yes
Cleaning sponge	No

with continuity testing and diode polarity testing can be very useful while tracing the circuit on a PCB board.

Left/right arrangement. It is almost impossible for a left-handed person to use the smart tweezers configured for right-handed persons. He will see the display upside down. So if you are left-handed, look for the configurability of the smart tweezer for left-hand use.

Some smart tweezers even come with angled tweezer spares, which are even more comfortable to work with SMD components.

Table 1 shows specifications of some smart tweezer models from different manufacturers.

Soldering tools

For electronics jobs involving a lot of rework and construction, you will need

to use some soldering tools quite frequently.

Soldering iron. Just like multimeters, soldering stations included in toolkits are not really of professional quality. Try to get at least a temperature-controlled soldering station. For good quality jobs you will need a professional quality temperature-controlled soldering station. To understand what to look for while buying a temperature-controlled soldering station, refer the buyers' guide published in EFM April 2013 issue.

De-soldering pump. A de-soldering pump, also known as solder sucker, is a manually-operated device that is used to remove solder from a printed circuit board. There are sophisticated de-soldering stations available but you cannot ever find them in a toolkit because they are



very expensive. De-soldering pump is very handy and does a reasonably good job when you are outside your lab to do some rework.

Solder. Solder is a metallic alloy which, when melted by the soldering iron, can join metallic surfaces together. The most common solder

Type	Tin (Sn)	Lead (Pb)	Silver (Ag)	Melting temperature
60/40	60%	40%	0%	183-190°C
63/37	63%	37%	0%	183°C
62/36/2	62%	36%	2%	179°C

